

§3.3 动量算符和角动量算符

一、动量本征函数及其归一化

1. 本征函数

动量算符

$$\vec{p} = -i\hbar\nabla$$

本征方程

$$-i\hbar\nabla\psi_{\vec{p}} = \vec{p}\psi_{\vec{p}} \quad (\vec{p} \text{ 是任何实矢量})$$

$$\begin{cases} -i\hbar\frac{\partial\psi}{\partial x} = p_x\psi \\ -i\hbar\frac{\partial\psi}{\partial y} = p_y\psi \\ -i\hbar\frac{\partial\psi}{\partial z} = p_z\psi \end{cases}$$

$$\psi_{\vec{p}}(\vec{r}) = C \exp\left[\frac{i}{\hbar}(p_x x + p_y y + p_z z)\right] = C \exp\left(\frac{i}{\hbar}\vec{p}\cdot\vec{r}\right)$$

p_x, p_y, p_z 可以取 $(-\infty, +\infty)$ 中连续变化的一切实数值

$$\int_{-\infty}^{+\infty} \left| \psi_{\vec{p}}(\vec{r}) \right|^2 d\vec{r} = |C|^2 \int_{-\infty}^{+\infty} d\vec{r} = \infty$$

在量子力学中, 坐标和动量的取值是连续变化的; 角动量的取值是离散的; 而能量的取值兼而有之, 则视边条件而定.

当然, 任何真实的波函数都不会是严格的平面波, 而是某种形式的波包 . 它只在空间某有限区域不为零.

如果此波包的广延比所讨论的问题中的特征长度大得多, 而粒子在此空间区域中各点的概率密度变化极微, 则不妨用平面波来近似描述其状态.

【注释】严格地说, 相应于连续谱的本征函数都在Hilbert空间之外, 但量子力学中仍然广泛使用连续谱的本征态作为基矢来展开. 从波函数的统计诠释来看, 我们可以把条件放松一些, 即不一定要要求本征函数都平方可积, 而只要求任何平方可积的波函数与它们的“标积”是有限值即可, 这不会对统计诠释造成困难.

● 函数归格化

$$\int_{\infty} \psi_{\vec{p}'}^*(\vec{r}) \psi_{\vec{p}}(\vec{r}) d\tau$$

$$= |C|^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \frac{i}{\hbar} \left[(p_x - p'_x)x + (p_y - p'_y)y + (p_z - p'_z)z \right] dx dy dz$$

$$= |C|^2 \int_{\infty} \exp \left[\frac{i}{\hbar} (p_x - p'_x)x \right] dx \int_{\infty} \exp \left[\frac{i}{\hbar} (p_y - p'_y)y \right] dy \int_{\infty} \exp \left[\frac{i}{\hbar} (p_z - p'_z)z \right] dz$$

$$= |C|^2 (2\pi\hbar)^3 \delta(p_x - p'_x) \delta(p_y - p'_y) \delta(p_z - p'_z)$$

$$= |C|^2 (2\pi\hbar)^3 \delta^3(\vec{p} - \vec{p}')$$

$$\psi_{\vec{p}}(\vec{r}) = \frac{1}{\sqrt{(2\pi\hbar)^3}} \exp \left(\frac{i}{\hbar} \vec{p} \cdot \vec{r} \right)$$

$$C = \frac{1}{\sqrt{(2\pi\hbar)^3}}$$

$$\int_{\infty} \psi_{\vec{p}'}^*(\vec{r}) \psi_{\vec{p}}(\vec{r}) d\tau = \delta^3(\vec{p} - \vec{p}')$$

● 箱归一化

先让粒子局限于有限空间 $[-L/2, L/2]$ 中运动(最后才让 $L \rightarrow \infty$)

此时, 为了保证动量算符为厄米算符, 就要求波函数满足周期性边条件.

$$C \exp\left[\frac{i}{\hbar}\left(-\frac{1}{2} p_x L + p_y y + p_z z\right)\right] = C \exp\left[\frac{i}{\hbar}\left(\frac{1}{2} p_x L + p_y y + p_z z\right)\right]$$

$$\exp\left(\frac{i}{\hbar} p_x L\right) = 1 \Rightarrow \frac{p_x}{\hbar} L = 2n_x \pi$$

$$\begin{cases} p_x = \frac{2\pi\hbar n_x}{L}, & (n_x = 0, \pm 1, \pm 2, \dots) \\ p_y = \frac{2\pi\hbar n_y}{L}, & (n_y = 0, \pm 1, \pm 2, \dots) \\ p_z = \frac{2\pi\hbar n_z}{L}, & (n_z = 0, \pm 1, \pm 2, \dots) \end{cases}$$

$$\Delta p_x = \Delta p_y = \Delta p_z = \frac{2\pi\hbar}{L} \propto \frac{1}{L}$$

 $L \rightarrow \infty$

本征值谱由分立谱变为连续谱

采用周期性边界条件后, 动量本征函数可以归一化

$$\psi_{\vec{p}} = \frac{1}{L^{3/2}} \exp\left(\frac{i}{\hbar} \vec{p} \cdot \vec{r}\right)$$

$$\int \psi_{\vec{p}}^* \psi_{\vec{p}} d\tau = \frac{1}{L^3} \int_{-L/2}^{L/2} dx \int_{-L/2}^{L/2} dy \int_{-L/2}^{L/2} dz = 1$$

二、角动量算符及其本征函数

1. 直角坐标表示

定义 $\vec{L} = \hat{\vec{r}} \times \vec{p} = -i\hbar \vec{r} \times \nabla$

$$\hat{\vec{L}} = \vec{r} \times \hat{\vec{p}} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & z \\ \hat{p}_x & \hat{p}_y & \hat{p}_z \end{vmatrix}$$

$$\hat{L}_x = y\hat{p}_z - z\hat{p}_y$$

$$\hat{L}_y = z\hat{p}_x - x\hat{p}_z$$

$$\hat{L}_z = x\hat{p}_y - y\hat{p}_x$$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

【思考1】 如何得到球坐标系下角动量 L_z 与 L^2 的表达式?

【思考2】 如何求解角动量 L_z 与 L^2 的本征值和本征函数?

【思考3】 如何应用角动量的本征值和本征函数?

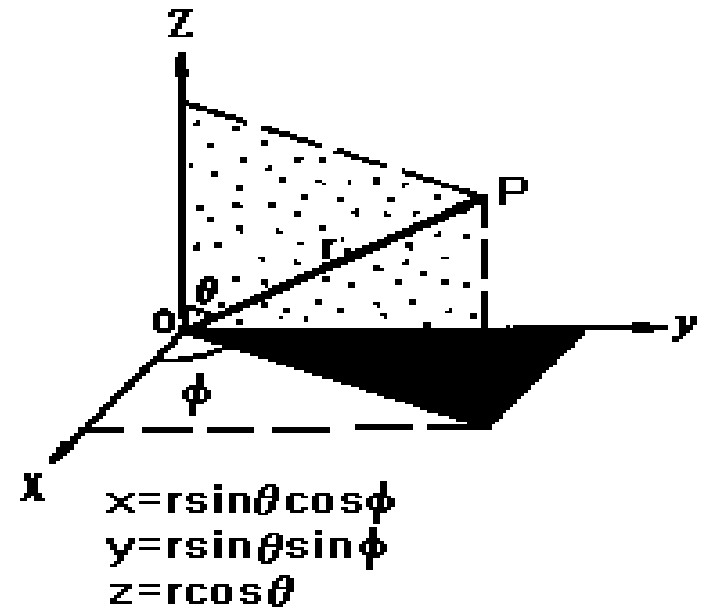
2. 球坐标表示

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta$$

$$r \in [0, \infty), \quad \theta \in [0, \pi], \quad \varphi \in [0, 2\pi)$$

$$\begin{cases} \hat{L}_x = i\hbar \left(\sin \varphi \frac{\partial}{\partial \theta} + \cot \theta \cos \varphi \frac{\partial}{\partial \varphi} \right) \\ \hat{L}_y = -i\hbar \left(\cos \varphi \frac{\partial}{\partial \theta} - \cot \theta \sin \varphi \frac{\partial}{\partial \varphi} \right) \\ \hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi} \end{cases}$$

$$L^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right]$$



3. L_z 的本征函数

本征方程 $L_z \psi = L_z \psi$

$$\frac{d\psi}{d\varphi} = \frac{i}{\hbar} L_z \psi(\varphi), \quad \psi(\varphi) = C \exp\left(\frac{i}{\hbar} L_z \varphi\right)$$

为了保证 L_z 的厄米性, 要求波函数满足周期性边界条件(或称为单值条件), 即:

$$\psi(\varphi + 2\pi) = \psi(\varphi)$$

【注释】周期性边界条件的讨论{参见曾谨言著《量子力学》卷I}

厄米性要求 $(\phi, L_z \psi) = (L_z \phi, \psi)$

$$(\phi, L_z \psi) = \int_0^{2\pi} \phi^*(\varphi) \frac{\hbar}{i} \frac{\partial}{\partial \varphi} \psi(\varphi) d\varphi$$

$$= \frac{\hbar}{i} \phi^*(\varphi) \psi(\varphi) \Big|_0^{2\pi} - \int_0^{2\pi} \frac{\hbar}{i} \left(\frac{\partial \phi^*(\varphi)}{\partial \varphi} \right) \psi(\varphi) d\varphi$$

$$= \frac{\hbar}{i} \phi^*(\varphi) \psi(\varphi) \Big|_0^{2\pi} + (L_z \phi, \psi)$$

$$\phi^*(2\pi) \psi(2\pi) - \phi^*(0) \psi(0) = 0 \quad \frac{\psi(2\pi)}{\psi(0)} = \frac{\phi^*(0)}{\phi^*(2\pi)}$$

此比值对所有波函数都是一样的

$$l'_z = 0, \quad \frac{\psi(2\pi)}{\psi(0)} = 1$$

因此对于任何一个波函数有

$$\psi(2\pi) = \psi(0)$$

—周期性条件

$$C e^{\frac{i}{\hbar} L_z \cdot 2\pi} = C$$

$$\cos \frac{1}{\hbar} L_z \cdot 2\pi = 1$$

$$\frac{1}{\hbar} L_z \cdot 2\pi = m \cdot 2\pi \quad (m = 0, \pm 1, \pm 2, \dots)$$

$$L_z = m\hbar$$

归一化 $\int_0^{2\pi} |\psi_m(\phi)|^2 d\phi = 1 \Rightarrow \text{取 } C = \frac{1}{\sqrt{2\pi}}$

$$\begin{cases} \Phi_m(\varphi) = \frac{1}{\sqrt{2\pi}} e^{im\varphi} \\ L_z = m\hbar, \quad m = 0, \pm 1, \pm 2, \dots \end{cases}$$

4. L^2 的本征函数

$$L^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right]$$

本征方程 $L^2 Y(\theta, \varphi) = \lambda \hbar^2 Y(\theta, \varphi)$

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y(\theta, \varphi)}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y(\theta, \varphi)}{\partial \varphi^2} = -\lambda Y(\theta, \varphi)$$

本征方程的求解: 设 $Y(\theta, \varphi) = \Theta(\theta) \exp(im\varphi)$

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta(\theta)}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} \Theta(\theta) = -\lambda \Theta(\theta), \quad 0 \leq \theta \leq \pi$$

 $\xi = \cos \theta, \quad \xi \in [-1, +1]$

$$\frac{d}{d\xi} \left[(1 - \xi^2) \frac{d\Theta(\xi)}{d\xi} \right] + \left(\lambda - \frac{m^2}{1 - \xi^2} \right) \Theta(\xi) = 0$$

$$\lambda = l(l+1) \quad (l = |m|, |m|+1, \dots)$$

$$\frac{d}{d\xi} \left[(1-\xi^2) \frac{dP_l^m(\xi)}{d\xi} \right] + \left(l(l+1) - \frac{m^2}{1-\xi^2} \right) P_l^m(\xi) = 0$$

$$\Downarrow \quad m = 0$$

$$\frac{d}{d\xi} \left[(1-\xi^2) \frac{dP_l(\xi)}{d\xi} \right] + l(l+1)P_l(\xi) = 0$$

Legendre多项式 $P_l(\xi) = \frac{1}{2^l l!} \frac{d^l}{d\xi^l} (\xi^2 - 1)^l$ **——Legendre方程**

缔合(或连带)Legendre函数

$$P_l^m(\xi) = \frac{1}{2^l l!} (1-\xi^2)^{m/2} \frac{d^{l+m}}{d\xi^{l+m}} (\xi^2 - 1)^l$$

L^2 的本征函数

$$Y_{lm}(\theta, \varphi) = N_{lm} P_l^m(\cos \theta) \exp(im\varphi)$$
$$l = 0, 1, 2, \dots; \quad m = l, l-1, \dots, -l$$

$$\int Y_{lm}^*(\theta, \varphi) Y_{lm}(\theta, \varphi) d\Omega = 1 \quad (d\Omega = \sin \theta d\theta d\varphi)$$

$$N_{lm} = (-1)^m \sqrt{\frac{(2l+1)}{4\pi} \frac{(l-m)!}{(l+m)!}}$$

头两阶球谐函数 ($l = 0, 1$)

$$Y_{00} = \sqrt{\frac{1}{4\pi}}$$

$$Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta$$

$$Y_{1,\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta \exp(\pm i\phi)$$

球谐函数的基本性质

$$\begin{cases} L^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm}, & l = 0, 1, 2, \dots \\ L_z Y_{lm} = m\hbar Y_{lm}, & m = l, l-1, \dots, -l \end{cases}$$

$$\int Y_{l'm'}^*(\theta, \varphi) Y_{lm}(\theta, \varphi) d\Omega = \delta_{l'l} \delta_{m'm}$$

$$Y_{lm}^*(\theta, \phi) = (-1)^m Y_{l,-m}(\theta, \phi)$$

【讨论】

- $Y_{lm}(\theta, \varphi)$ 是 L^2 和 $\hat{L}_z$ 的共同本征函数
- l 为轨道角量子数; m 称为磁量子数
 $l = 0, 1, 2, \dots$ 分别为s、p、d、f、...态
- L^2 的本征值是 $f_l = 2l+1$ 度简并的.
- 算符集合 $\{L^2, \hat{L}_z\}$ 是描述转动的力学量完全集, 用量子数 $\{l, m\}$ 可以完全确定转动态.

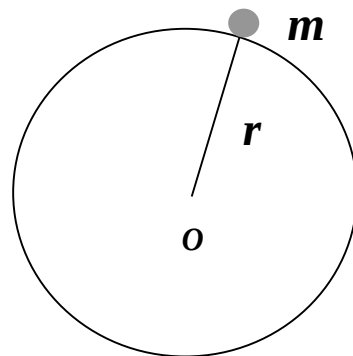
【例题1】 粒子沿光滑球面自由运动(忽略重力势能), 求能量本征值和本征波函数.

【解答】

$$\hat{H} = \frac{\hat{L}^2}{2I}, \quad I = mr^2$$

力学量完全集 $\{L^2, L_z\}$

共同本征函数: 球谐函数



$$\hat{H}Y_{lm}(\theta, \varphi) = \frac{\hat{L}^2}{2I}Y_{lm}(\theta, \varphi) = \frac{l(l+1)\hbar^2}{2I}Y_{lm}(\theta, \varphi)$$

$$\hat{L}_z Y_{lm}(\theta, \varphi) = m\hbar Y_{lm}(\theta, \varphi)$$

能量本征值

$$E_l = \frac{l(l+1)\hbar^2}{2I}, \quad l=0, 1, 2, \dots$$

本征波函数

$$\Psi(\theta, \phi) = Y_{lm}(\theta, \phi), \quad (l=0, 1, 2, \dots; m=-l, -l+1, \dots, l-1, l)$$

能量的简并度

$$f_l = 2l + 1$$

【例题2】一量子体系处于 $\psi(\theta, \varphi) = \frac{1}{2}Y_{10} - \frac{\sqrt{3}}{2}Y_{1-1}$ 态中, 问角动量 $\hat{L}^2$ 、 $\hat{L}_z$ 力学量的可能值各为多少? 概率多大?

【解答】

$$\hat{L}^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm}$$

$$\hat{L}_z Y_{lm} = m\hbar Y_{lm}$$

	可能値	概率	平均值
$\hat{L}^2$	$2\hbar^2$	1/4	$2\hbar^2$
	$2\hbar^2$	3/4	
$\hat{L}_z$	0	1/4	$-\frac{3}{4}\hbar$
	$-\hbar$	3/4	